

CHAPTER 29 — MAGNETIC FIELDS DUE TO CURRENTS

Quick-Revision Notes | 4 Topics Covered

TOPIC 1 — BIOT–SAVART LAW (Section 29-2)

Key Definition

A moving charged particle produces a magnetic field around itself. A current of moving charges therefore produces a magnetic field around the wire. This is the basis of all electromagnetic devices.

The differential magnetic field dB produced at point P by a current-length element $i ds$ is given by the Biot–Savart Law.

Biot–Savart Law — Formulas

Magnitude (Eq. 29-1)	$dB = (\mu_0/4\pi) \cdot (i ds \sin\theta) / r^2$	θ = angle between ds and $\hat{r}$
Vector Form (Eq. 29-3)	$d\vec{B} = (\mu_0/4\pi) \cdot (i d\vec{s} \times \hat{r}) / r^2$	Biot–Savart Law
Long Straight Wire (Eq. 29-4)	$B = \mu_0 i / (2\pi R)$	R = perp. distance from wire
Semi-infinite Wire (Eq. 29-7)	$B = \mu_0 i / (4\pi R)$	Wire starting at P's level
Circular Arc (Eq. 29-9)	$B = \mu_0 i \phi / (4\pi R)$	ϕ = arc angle in radians
Full Circle	$B = \mu_0 i / (2R)$	$\phi = 2\pi$

Right-Hand Rule

- Grasp the wire with right hand, extended thumb pointing in direction of current.
- Curled fingers show direction of magnetic field lines (concentric circles around wire).

Key Points

- Magnetic field lines around a long straight wire are concentric circles.

- Superposition applies: net B = vector sum of fields from all current elements.
- Unlike electric fields (scalar source dq), magnetic sources $i \, d\vec{s}$ are vectors — making integration more complex.
- For straight sections pointing at P , $\sin \theta = 0 \rightarrow dB = 0$ (no contribution).

Proof of Eq. 29-4 (Long Straight Wire) — Summary

- Use substitutions: $\sin\theta = R / (S^2 + R^2)^{1/2}$ and $r^2 = S^2 + R^2$
- Integrate using integral-19: $\int dx/(x^2+a^2)^{3/2} = x / [a^2(x^2+a^2)^{1/2}]$
- Evaluate from 0 to ∞ : result = 1, giving $B = \mu_0 i / (2\pi R)$

TOPIC 2 — FORCE BETWEEN TWO PARALLEL CURRENTS

(Section 29-2 continued)

Key Concept

When two parallel wires carry currents, each wire lies in the magnetic field produced by the other wire, and so each wire experiences a force.

Force Formula

Force on wire b (Eq.)	$F_{ba} = \mu_0 L i_a i_b / (2\pi d)$	L = length, d = separation
Vector form	$\vec{F}_{ba} = i_b L \vec{r} \times \vec{B}_a$	Cross product gives direction

Attraction / Repulsion Rule

- Currents in SAME direction → wires ATTRACT each other.
- Currents in OPPOSITE directions → wires REPEL each other.

Rail Gun — Application

- Large current travels along one rail, crosses a conducting fuse, and returns along the other rail.
- The fuse vaporizes immediately, forming a conducting gas.
- Magnetic field between rails exerts a force on the gas, propelling the projectile.
- Acceleration can reach 5×10^6 g; launch speed ~10 km/s within 1 ms.

Sample Problem Values (Two parallel wires)

Given: $i_1 = 15$ A, $i_2 = 32$ A, $d = 5.3$ cm. Base angles = 45° .

$$B = [\mu_0 / (2\pi d \cos 45^\circ)] \times \sqrt{(i_1^2 + i_2^2)} = 1.89 \times 10^{-4} \text{ T} \approx 190 \text{ } \mu\text{T}$$

$$\text{Angle } \phi \text{ of net B from } B_2 \text{ direction: } \phi = \tan^{-1}(i_1/i_2) = \tan^{-1}(15/32) = 25^\circ$$

TOPIC 3 — AMPERE'S LAW (Section 29-3)

Statement

The line integral of the magnetic field around any closed loop (Amperian loop) equals μ_0 times the net current enclosed by that loop.

Ampere's Law	$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$	Closed loop integral
--------------	---	----------------------

Sign Convention (Right-Hand Rule for Currents)

- Curl right hand around the Amperian loop with fingers in direction of integration.
- Current in direction of outstretched thumb $\rightarrow$ positive (+).
- Current in opposite direction $\rightarrow$ negative (-).
- Example: $i_{\text{enc}} = i_1 - i_2$ if i_1 is positive and i_2 is negative.

Magnetic Field Outside a Long Straight Wire

Choose circular Amperian loop of radius r centered on the wire ($r > R$).

Outside wire ($r > R$)	$B = \mu_0 i / (2\pi r)$	Same as Biot-Savart result
--------------------------	--------------------------	----------------------------

Magnetic Field Inside a Long Straight Wire

Current is uniformly distributed. Only encircled current contributes.

$i_{\text{enc}} = i \times (\pi r^2)/(\pi R^2)$ — proportional to area

Inside wire ($r < R$)	$B = (\mu_0 i / 2\pi R^2) \cdot r$	B increases linearly with r
-------------------------	------------------------------------	-------------------------------

Sample Problem — Hollow Cylinder

Cylinder: inner radius $a = 2.0$ cm, outer radius $b = 4.0$ cm, $J = cr^2$ ($c = 3.0 \times 10^6$ A/m⁴).

At $r = 3.0$ cm: $i_{\text{enc}} = \pi c(r^4 - a^4)/2$

Result: $B = 2.0 \times 10^{-5}$ T

Key Points

- Ampere's Law works best when the field has high symmetry (straight wire, toroid, solenoid).
- Only the NET current piercing the Amperian loop surface matters.
- Currents outside the Amperian loop contribute zero to the integral (they cancel out).

TOPIC 4 — SOLENOIDS AND TOROIDS (Section 29-4)

Solenoid

A solenoid is a long, tightly wound helical coil of wire. When length $\gg$ diameter, the interior field is strong and uniform; the exterior field is nearly zero.

Ideal Solenoid (Eq. 29-23)	$B = \mu_0 i n$	$n = \text{turns per unit length (N/L)}$
----------------------------	-----------------	--

- The rectangular Amperian loop encloses nh turns $\rightarrow i_{\text{enc}} = i(nh)$.
- Three sides of the loop contribute zero ($B=0$ outside, or $B \perp ds$); only the inner side ab contributes.
- Result: $Bh = \mu_0 i nh \rightarrow B = \mu_0 i n$

Solenoid Sample Problem

$L = 1.23 \text{ m}$, $d = 3.55 \text{ cm}$, $i = 5.57 \text{ A}$, 5 layers $\times$ 850 turns each.

$n = 5 \times 850 / 1.23 \rightarrow B = \mu_0 i n = (4\pi \times 10^{-7})(5.57)(5 \times 850 / 1.23) = 24.2 \text{ mT}$

Toroid

A toroid is a solenoid bent into a donut (torus) shape. The magnetic field is confined inside the toroid and is not uniform (varies with r).

Toroid Field (Eq. 29-24)	$B = (\mu_0 i N) / (2\pi r)$	$N = \text{total turns, } r = \text{radius to point}$
--------------------------	------------------------------	---

- Outside the toroid (or in the central hole): $B = 0$ (encircled current sums to zero).
- Field is stronger near the inner edge than the outer edge.

Comparison Table

Device	Formula	Field Location	Uniformity
Long Wire	$B = \mu_0 i / 2\pi R$	Everywhere outside	Not uniform
Solenoid	$B = \mu_0 i n$	Inside only	Uniform
Toroid	$B = \mu_0 i N / 2\pi r$	Inside torus only	Not uniform ($1/r$)

MASTER FORMULA SHEET — Chapter 29

Configuration	Formula	Variables
Biot–Savart (scalar)	$dB = \mu_0 i \, ds \sin\theta / (4\pi r^2)$	θ = angle between ds and $\hat{r}$
Biot–Savart (vector)	$d\vec{B} = \mu_0 (i \, d\vec{s} \times \hat{r}) / (4\pi r^2)$	Vector cross product
Long straight wire	$B = \mu_0 i / (2\pi R)$	R = perpendicular distance
Semi-infinite wire	$B = \mu_0 i / (4\pi R)$	Wire from 0 to ∞
Circular arc	$B = \mu_0 i \phi / (4\pi R)$	ϕ in radians, at center C
Full circle	$B = \mu_0 i / (2R)$	$\phi = 2\pi$
Force: parallel wires	$F/L = \mu_0 i_a i_b / (2\pi d)$	d = separation
Ampere's Law	$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$	Closed loop
Wire outside (Ampere)	$B = \mu_0 i / (2\pi r)$	$r > R$
Wire inside (Ampere)	$B = (\mu_0 i / 2\pi R^2) r$	$r < R$
Ideal solenoid	$B = \mu_0 i n$	$n = N/L$ (turns/meter)
Toroid	$B = \mu_0 i N / (2\pi r)$	N = total turns

$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A} \text{ (permeability of free space)}$$
